Basic mechanical ventilator in the critical care

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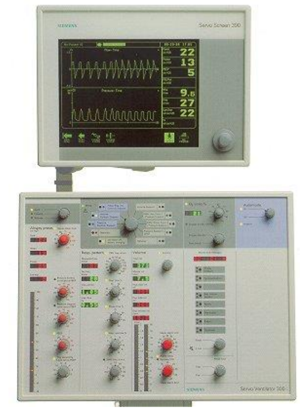
Outline:

- Introduction.
- Definition.
- Indications.
- Type of mechanical ventilatos.
- Ventilator parameters.
- Modes of ventilation.
- Ventilator Settings.
- Weaning.
- Complications of Mechanical Ventilation.

Definition:

- *Mechanical ventilation is the use of a mechanical device (machine) to inflate and deflate the lungs.*
- *A mechanical ventilator is simply a machine used to replace or supplement the natural function of breathing*



Siemens Servo 300

Siemens Servo 900C

Drager Evita 4

Indications of Mechanical Ventilation

- **Apnea : absence of breathing**(e.g., neuromuscular or cardiopulmonary collapse)
- **Acute ventilatory failure**, which is generally defined as a pH of less than or equal to 7.25 with ($PaCO_2$) greater than or equal to 50 mm Hg.
- **Severe hypoxemia:** (PaO_2) of less than or equal to 50 mm Hg on room air indicates a critical level of oxygen in the blood.
- **Respiratory muscle fatigue:e.g.,** chronic lung disease are examples COPD.

Types of Mechanical ventilators:

- Negative pressure ventilator
  :
  iron lung.
- Positive pressure ventilator
  :
  modern ventilator

Negative pressure ventilator

Positive pressure ventilator

- Improve pulmonary gas exchange
- Reverse hypoxemia
- Reverse acute respiratory acidosis
- Relieve respiratory distress
- Reverse respiratory muscle fatigue
- Prevent and reverse atelectasis
- Improve lung compliance

Criteria for institution of ventilatory support:

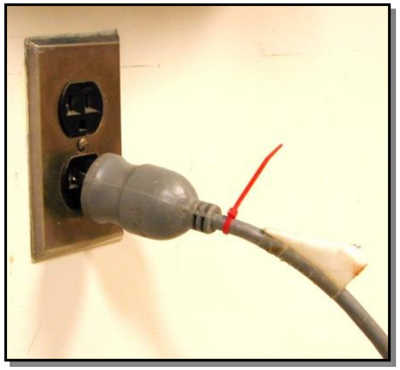
Parameters	Ventilation indicated	Normal range
Pulmonary function studies:		
Respiratory rate (breaths/min).	> 35	10-20
Tidal volume (ml/kg body wt)	< 5	5-7
Vital capacity (ml/kg body wt)	< 15	65-75
Maximum Inspiratory Force (cm HO_2)	<-20	75-100
Arterial blood Gases		
PH	< 7.25	7.35-7.45
PaO_2 (mmHg)	< 60	75-100
$PaCO_2$ (mmHg)	> 50	35-45

Ventilator classification

- Power input
- Power transmission or conversion
- Control scheme

Power input

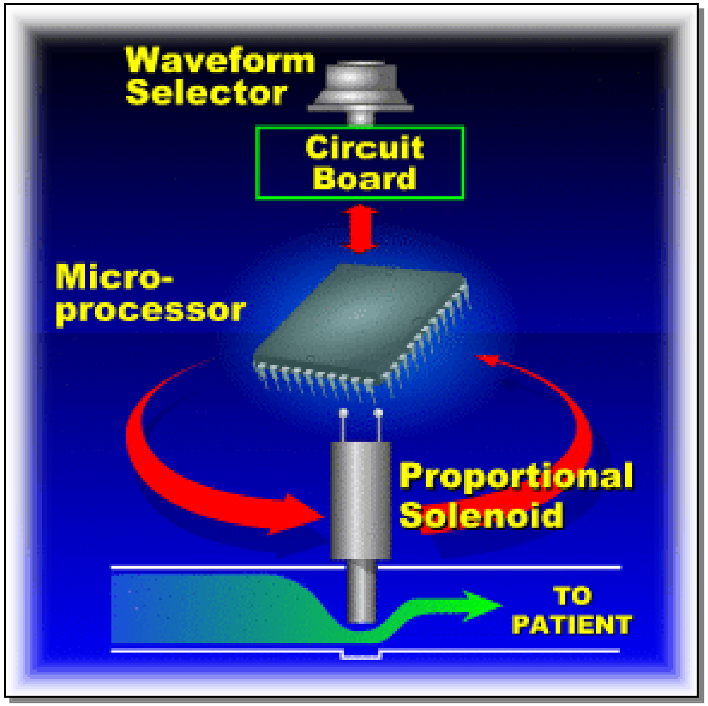
- Electric
  - AC
  - DC (battery)
    - internal
    - External



- Pneumatic
  - Oxygen (tanks or wall)
  - Compressed air
    - Internal air compressor
    - External air compressor
    - Internal turbine

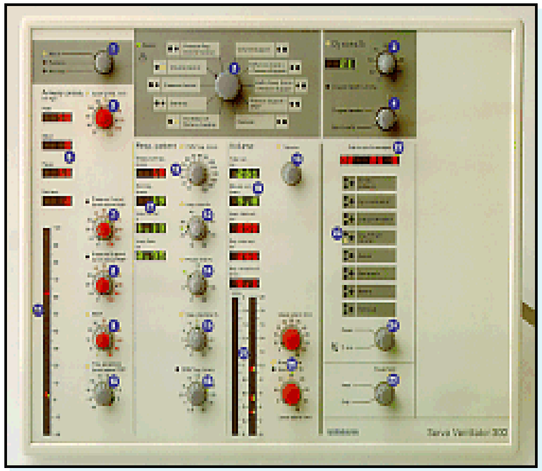
Control scheme

- **Control variables & waveform selection:**
  - Pressure
  - Volume
  - Flow
  - Time



Control scheme

- **Phase variables:**
  - Trigger variable
  - Limit variable
  - Cycle variable
  - Baseline variable



The patient circuit

- Is the interface between the ventilator and the patient.
- It has three major components, the inspiratory limb, the patient way and the expiratory limb.

Inspiratory Limb:

- The inspiratory limb provides a passage for blended gas to be delivered to the patient from the ventilator.
- There are three components associated with this limb of patient circuit. The inhalation filter, the humidifier and the nebulizer.
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DH-100C1

Inhalation Filter:

- This provides filtration of blended gas inspired by the patient from the ventilator.
- It also protects the machine from infectious materials when the patient exhales.

Humidifiers:

Humidifiers are employed in res
therapy to:

1) supply enough water vapor to the inspired gas to make it comfortable for the patient
2) heating the gas to provide 100% relative humidity at body temperature.

- To accomplish humidification three factors are important.

1) **Time of contact between the gas and water.**
2) **Surface area involved in gas/water contact.**
3) **Temperature.**

Classification of Ventilators

Ventilators are categorized as either

- **Negative pressure**
- **Positive pressure**.

Negative Pressure Ventilators

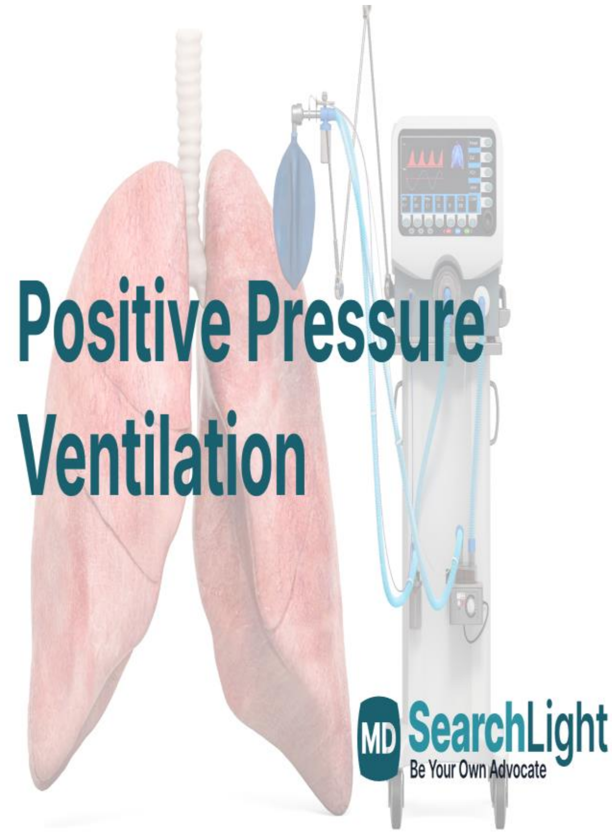
- Negative-pressure ventilators exert a negative pressure on the external chest.
- Decreasing the intrathoracic pressure during inspiration allows air to flow into the lung. Physiologically, this type of assisted ventilation is similar to spontaneous ventilation.

Negative Pressure Ventilators

- It is used in chronic respiratory failure associated with neuromuscular conditions, such as poliomyelitis, muscular dystrophy, and myasthenia gravis.
- Negative- pressure ventilators do not require intubation of the airway.
- There are several types of negative pressure ventilators: iron lungs, and chest cuirass.

Positive Pressure Ventilation

- Positive-pressure ventilators inflate the lungs by exerting positive pressure on the airway, pushing air in, forcing the alveoli to expand during inspiration. Expiration occurs passively.



Positive Pressure Ventilation

1- **Invasive Mechanical Ventilation**: Endotracheal intubation or tracheostomy is usually necessary for Positive-pressure ventilators.

2- Noninvasive positive-pressure ventilation does not require intubation.

Positive-pressure modes of ventilation have traditionally been categorized into

- **Volume ventilation**
- **Pressure ventilation**

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Volume Ventilation

- Volume ventilation has traditionally been the most popular form of PPV, largely because **Tidal Volume (Vt) and Minute Ventilation (MV)** are ensured, which is an essential goal in the patient with acute illness.
- The preset volume of air delivered with each inspiration. Once this preset volume is delivered to the patient, the ventilator cycles off and exhalation occurs passively.
- A preset Vt is delivered with each breath regardless of resistance and compliance. Tidal Volume is stable from breath to breath, but airway pressure may vary.

Volume Modes

1. ***Control Ventilation (CV) or Controlled Mandatory Ventilation (CMV)***
2. ***Assist/Control (A/C) or Assisted Mandatory Ventilation (AMV***
3. ***Synchronized Intermittent Mandatory Ventilation (SIMV)***

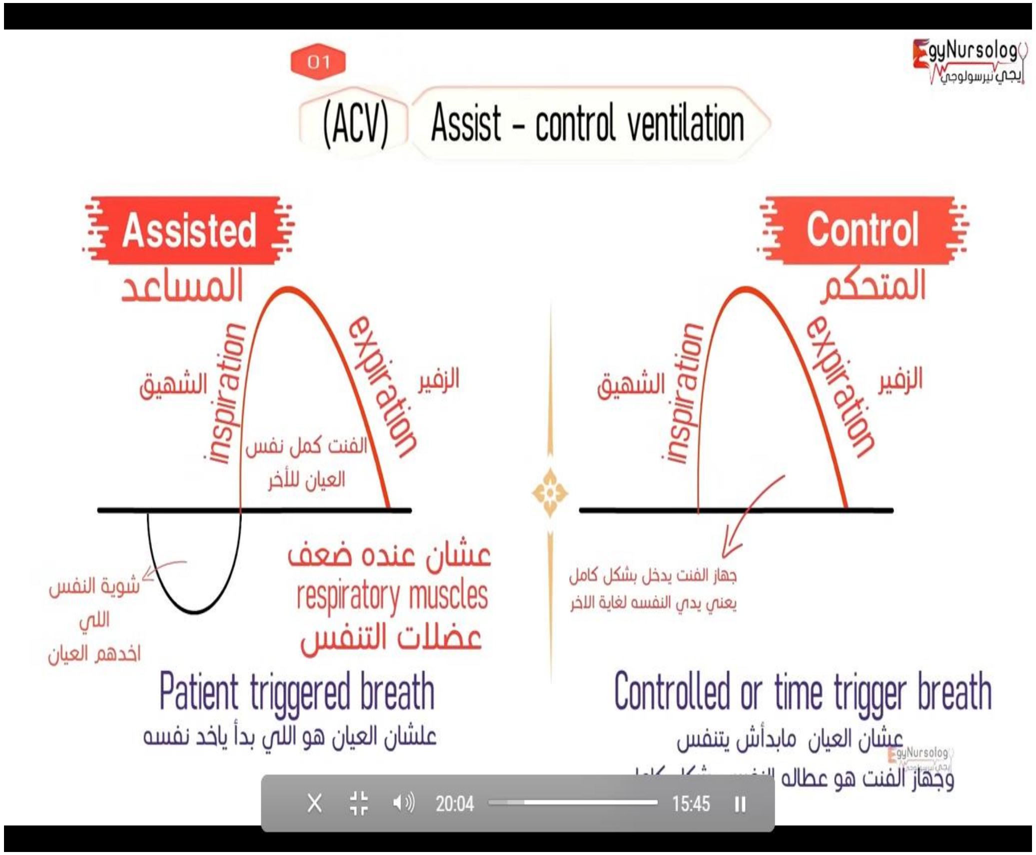
Volume Modes

- ***Control Ventilation (CV) or Controlled Mandatory Ventilation (CMV)***

  With this mode, the ventilator provides all the patient’s minute ventilation. The clinician sets the **respiratory rate, Vt, inspiratory time, and positive End Expiratory Pressure (PEEP).**

Volume Modes

- ***Assist/Control (A/C) or Assisted Mandatory Ventilation (AMV)*** When the patient initiates a spontaneous breath. The ventilator sensitivity also is set, and a full-volume breath is delivered. This option requires that a rate, Vt, inspiratory time, and PEEP be set for the patient.



Volume Modes

- ***Synchronized Intermittent Mandatory Ventilation (SIMV)***

This mode requires that rate, Vt, inspiratory time, sensitivity, and PEEP are set by the clinician. In between mandatory breaths, patients can spontaneously breathe at their own rates and Vt. With SIMV, the ventilator synchronizes the mandatory breaths with the patient's own inspirations.

Pressure Ventilation

- The clinician selects the desired pressure level and the Vt is determined by the selected pressure level, resistance, and compliance.

Types Pressure mode

1. ***Pressure Support Ventilation (PSV)***
2. ***Pressure-Controlled/Inverse Ratio Ventilation (PC/IRV)***
3. **Non-invasive Bilateral Positive-Pressure Ventilation Mode(BiPAP)**
4. ***Continuous Positive Airway Pressure (CPAP)***

Pressure Modes

1- Pressure-Controlled

- Which delivers a preset pressure for a set time

2- Pressure Support Ventilation (PSV)

- Patient **initiates** each breath
- Ventilator **assists** with a fixed pressure level
- Patient controls **rate**, **timing**, and **tidal volume**

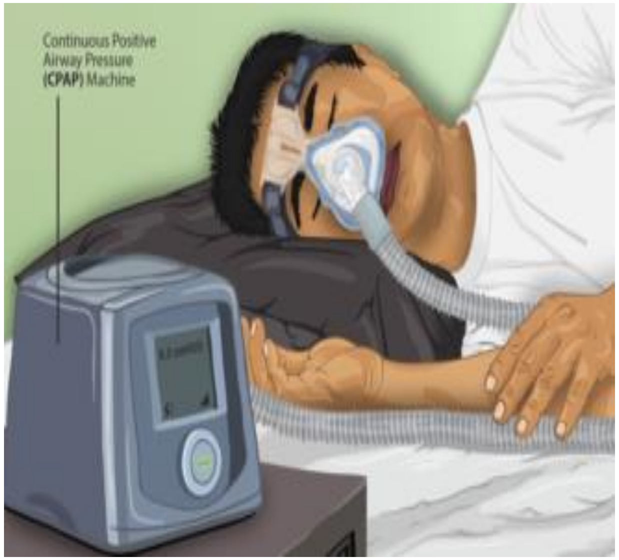
It's commonly used for:

- Weaning from mechanical ventilation
- Providing comfort while maintaining some ventilatory assistance

- **BiPAP** is beneficial in patients with worsening hypoventilation, obstructive apneic episodes, or both.
- It is also useful to prevent intubation in patients with respiratory failure and hypercarbia as well as to prevent reintubation following extubation in borderline cases.
- Use of a full face mask may increase the risk for aspiration and risk for re-breathing carbon dioxide; therefore, ventilation with a full face mask should be used cautiously.
- Thick or profuse secretions and poor cough may be relative contraindications for BiPAP.

Continuous Positive Airway Pressure (CPAP)

- Continuous positive airway pressure (CPAP) provides positive pressure to the airways throughout the respiratory cycle.
- CPAP assists spontaneously breathing patients to improve their oxygenation by elevating the end-expiratory pressure in the lungs keeping the upper airway and trachea open throughout the respiratory cycle.
- CPAP can be used for intubated and non-intubated patients.



To use CPAP, the patient must be breathing independently.

Ventilator Settings

- **Tidal Volume (V_T)**
  - amount of gas delivered with each preset breath
  - in mechanically ventilated patients it's usually set at 6 – 8 ml/kg
- **Respiratory Rate (RR)**
  - the frequency of breaths delivered by the ventilator
- **Fraction of Inspired Oxygen (FIO_2)**
  - the fraction of inspired oxygen delivered to the patient by the ventilator
  - change by ABG and O_2 saturation

Ventilator Settings

- **Inspiratory : Expiratory Ratio (I : E Ratio)**
  - usually set at 1 : 2, may be manipulated to facilitate gas exchange
- **Peak Inspiratory Pressure (PIP)**

  Measured in cmH_2O, this parameter controls the maximum inspiratory pressure to be delivered to the patient during a pressure controlled machine breath.
- **Adjuncts to Mechanical Ventilation**
  - PEEP, CPAP, PSV

Ventilator Settings

- **Pressure Limits**
  - high pressure limit is the maximum pressure the ventilator can generate to deliver the preset V_T
  - usually set 10 – 20 cm H_2O above the PIP
- **Apnea:**

  Apnea ventilation is **initiated when the preset apnea alarm is triggered.** The Ventilator will begin to deliver controlled breaths at the operator selected settings.
- Apnea ventilation **is terminated when the patient begins to breathe spontaneously** or a manual breath is delivered to the patient.

Ventilator Settings

Minute Volume:

Represents the **patient's exhaled tidal volume** (mechanical and spontaneous) over time.

Peak Pressure

Indicates the peak inspiratory pressure achieved during the last delivered breath.

VENTILATOR ALARMS MUST NEVER BE IGNORED OR DISARMED!!!!

- Loss of Power
- ELECTRICAL FAILURE ALARMS ARE A MUST FOR ALL VENTILATORS
  - Most ICU ventilators do not have battery back up like at home
  - Know back up system (special emergency outlets)
  - Check that power cord plug not accidentally disconnected

Alarms:

- **Frequency**
  - Alarms if RR goes above or below set levels
- **Volume**
  - Volumes go above or below preset levels (ie. VT or minute volume)
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Pressure

- Change in inspiratory or peak airway pressure above or below preset limits

Low Pressure Alarms	High Pressure Alarms
Disconnection	↓ Compliance: secretions
Loss of V_T	pneumothorax
Leak	ARDS/Pulmonary Edema
extubation	bronchospasm
	ETT in R mainstem
	"bucking"
	coughing
	pt. biting on tube
	tubing kinked
	H_2O in tubing

Monitored parameters

- Spontaneous VT
- Vital Capacity (VC)
- Negative Inspiratory Force (NIF)
- Compliance
- Minute Volume (MV)
  - determines alveolar ventilation (RR x VT = MV)
- Airway Placement / Patency
- ABGs

Weaning parameters:

- *Some of the objective parameters*

(a) PaO_2/FiO_2 ratio >150–200;

(b) Level of positive end expiratory pressure (PEEP) between 5–8 cm H_2O;

(c) FiO_2 level <50%;

(d) pH > 7.25;

(e) ability to initiate spontaneous breaths.

Some of the subjective parameters

- (a) Hemodynamic stability;
- (b) Absence of active myocardial ischemia;
- (c) Absence of clinically significant, vasopressor-requiring hypotension;
- (d) Appropriate neurological examination;
- (e) Improving or normal appearing chest radiogram;
- (f) Adequate muscular strength allowing the capability to initiate/sustain the respiratory effort.

SPONTANEOUS BREATHING TRIALS

multi trials, and many studies failed to demonstrate any significant differences between:

(a)continuous positive airway pressure of 5 cm H_2O versus T-piece for one hour

(b) T-piece versus pressure support of 7 cm H_2O

(c) duration of spontaneous breathing trial of 30 minutes versus 120 minutes.

Indicator of failure during spontaneous breathing trials:

Inadequate gas exchange

- Arterial oxygenation saturation (SaO_2) <85% – 90%
- PaO_2 <50 – 60 mmHg
- pH < 7.32
- Increase in $PaCO_2$ >50 mmHg

Unstable ventilatory/respiratory pattern

- Respiratory rate >30 – 35 breaths/minute
- Respiratory rate change over 50%

Hemodynamic instability

- Heart rate >120 – 140 beats/minute
- Heart rate change greater than 20%
- Systolic blood pressure >180 mmHg or <90 mmHg
- Blood pressure change greater than 20%
- Vasopressors required

Change in mental status

- Coma
- Agitation
- Anxiety

Signs of increased work of breathing

- Nasal flaring
- Paradoxical breathing movements
- Use of accessory respiratory muscles

Onset of worsening discomfort ± diaphoresis

Complications of Mechanical Ventilation

- ***Barotrauma***
  - presence of extra alveolar air
  - this air may escape (usually due to alveolar or bleb rupture) into the:
    - pleura (pneumothorax)
    - mediastinum (pneumomediastinum)
    - pericardium (pneumopericardium)
    - under the skin (subcutaneous emphysema or crepitus)

❑ Cardiovascular

- decreased venous return and CO
- may be manifested by decreased BP, decreased LOC, decreased urinary output, weak pulses, fatigue

❑ Gastrointestinal

- stress ulcers
- GI bleeding
- distention
- paralytic ileus
- Malnutrition

- **Acid – Base Disturbances**
  - respiratory alkalosis versus respiratory acidosis

- **O_2 Toxicity**
  - occurs with high concentrations of oxygen (FIO_2 >60%)

- **Aspiration**
  - gastric distention, impaired gastric emptying and esophageal reflux predispose patient to aspiration

Infection

- patients with artificial airways are at increased risk for pulmonary infection
- ETT suctioning can also predispose the patient to infection and frequently cause nosocomial infection
- Intubated patients have VAP

Immobility

- complications: muscle weakness/wasting, contractures, loss of skin integrity, pneumonia, DVT $\rightarrow$ PE, constipation, ileus

Psychological Complications

- patient may experience stress/anxiety due to being on a machine to breathe
- communication becomes challenging
- loses autonomy/control over care
- altered sleep pattern may occur
- depression may occur
- may develop psychological dependence on vent even if can physically be weaned

ANY QUESTION

THANK YOU